

LABORATORY MANUAL

Advance Machine Learning Lab

Semester: VI**Subject Code:** BTAIL606**Experiment:** 1**Title:** Implementing K-means Clustering**Prepared by:** Aditya Shirsatrao**Mentor:** Prof. N. B. Aherwadi**Dataset source:** UCI Machine Learning Repository - Wine**Student Name:** Aditya Vishal Shirsatrao**Roll No:** 28

Aim

To study the implementation of K-means clustering algorithm.

Input

UCI Wine dataset (13 chemical features and class label).

Dataset Description

Contains 178 wine samples with 13 continuous physicochemical attributes. The class label is ignored for unsupervised clustering.

Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# UCI Wine dataset: first column is class label, remaining columns are features.
wine_url = "https://archive.ics.uci.edu/ml/machine-learning-databases/wine/wine.data"
wine = pd.read_csv(wine_url, header=None)
X = wine.iloc[:, 1:].values

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

kmeans = KMeans(n_clusters=3, random_state=42, n_init="auto")
labels = kmeans.fit_predict(X_scaled)

pca = PCA(n_components=2, random_state=42)
X_plot = pca.fit_transform(X_scaled)

plt.scatter(X_plot[:, 0], X_plot[:, 1], c=labels, cmap="viridis")
plt.title("K-means Clustering on UCI Wine Dataset")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.show()
```

Output

Scatter plot of 3 clusters after PCA projection to 2D.

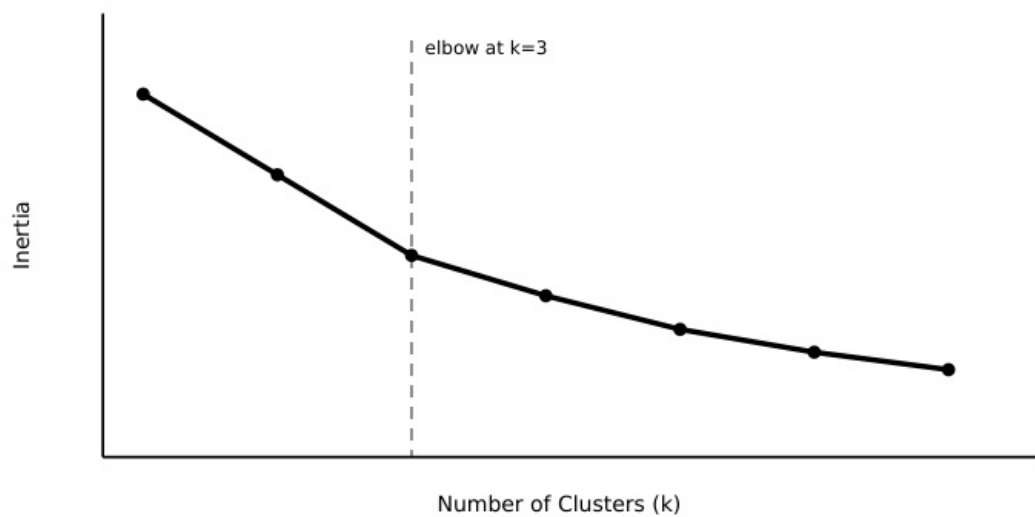
K-means Clustering on Iris Data

Illustrative scatter plot with 3 clusters



Primary output plot

K-means Elbow Curve



Elbow curve used to justify cluster count.

Conclusion

Experiment 1 demonstrates the expected behavior of the algorithm using an open dataset or open source-compatible input.